Development Pipeline Design

2026-10-21

Overview

This pipeline takes one development request around a full loop — design · implementation · testing · documentation · commit — then stops and waits for the next request.

It solves two problems: the design slowly eroding when requests are handled at face value, and working context living only in the conversation and vanishing by the next request.

At its core it is a single loop.

```
read context → order → execute → evaluate → limitations → update context → repeat
```

Step	Meaning
Order	Grasp the underlying intent of the request and turn it into an extensible design
Execute	Change the code
Evaluate	Write and run tests
Limitations	Analyze the whole run and surface unresolved items
Update context	Update the spec documents + commit

Five agents split these steps, each reading only its own share in an independent context; a sixth (F) keeps the context docs compact.

Core principles

Three principles explain every other design decision.

Limitations become the next order. The loop's final output isn't code but a list of unresolved items. When the next request arrives, A reads this first and designs with it and the request together. This closes the loop and makes it a structure that can run even without outside requests.

Documents are the state. There is no separate context store. The files in `.dev/context/` are themselves the project's state; the loop starts by reading them and ends by updating them. Updating documents isn't a side task — it's what closes the loop.

Context independence is token efficiency. If agents share full conversation transcripts, cost grows quadratically as the loop gets longer. A splits its analysis into order files, and B · C · D read only their own. That's why they run independently.

Directory structure

```
.dev/
├─── context/                persistent — the project's state
│    ├─── 00-project.md        accounts · repos · environment (secrets by reference only,
never values)
│    ├─── 01-features.md       feature spec
│    ├─── 02-design.md         visual design
│    ├─── 03-architecture.md   code architecture
│    └─── 04-devlog.md         development context (limitations / optimization targets)
└─── loops/                   volatile — handoff between agents
     └─── 2026-10-21-001/
          ├─── order-B.md       A → B
          ├─── order-C.md       A → C
          └─── result.md        B diff summary + C test results
```

`loops/` is the key to token efficiency. Orders have to be split into files so each agent reads only its own. Only the last 10 are kept; the rest are deleted automatically. Their content survives as summaries in commit messages and the devlog.

Document system

Specs are split along three axes. The rule: **if the reason to change differs, it's a different document.**

Document	Question	Content	Changes when
01-features	What can it do	use cases, inputs/outputs, state transitions, exceptions	what users want changes
02-design	How does it look	layout, tokens, components, interactions	how it looks changes
03-architecture	How does it work	module structure, data flow, dependencies, interfaces	how it's built changes

Boundary rule: "it happens" is a feature, "it looks like this" is design, "it's built this way" is architecture. Items that straddle are split and written on each side.

Cross-references: tie IDs together like `F-012 ↔ D-07 ↔ U-23`. Without them, the three documents age separately and drift apart.

04-devlog splits into two sections. Routing diverges here, so always record them separately.

```
## Limitations / Improvements   ← consumed by A (changes behavior)
- token refresh not implemented (F-031)
- no retry on upload failure

## Optimization Targets         ← consumed by E (behavior-preserving)
- duplicated validation logic in AuthProvider · SessionStore
- parseConfig() is 180 lines with 4 responsibilities
```

When unsure, send it up (to A). A second look by A is cheaper than E overstepping.

Agents

Each agent has an independent context. They are implemented as subagents.

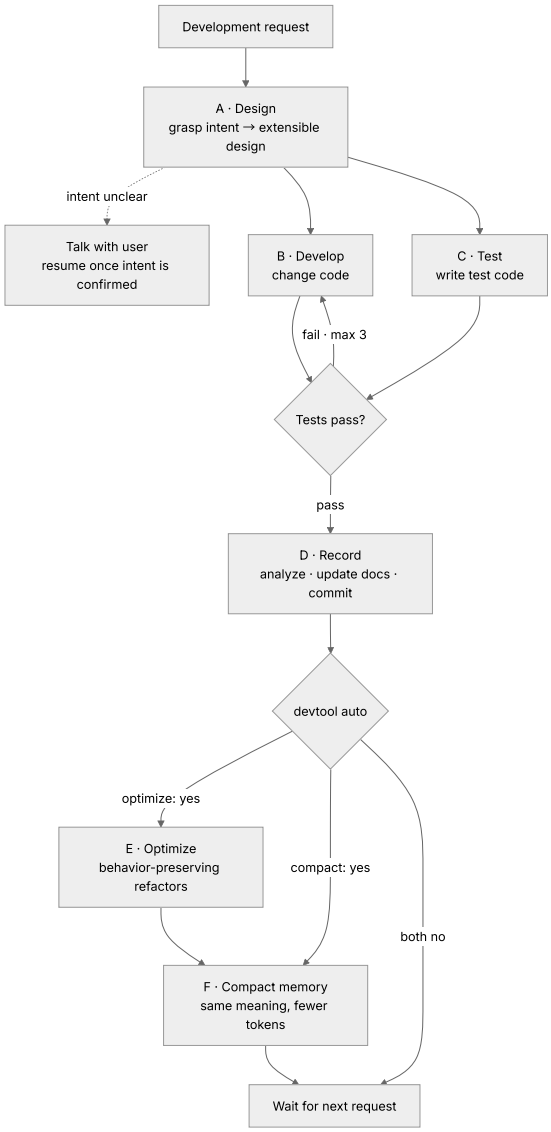
	Role	Reads	Writes	Commits
A	grasp intent → turn into extensible design, distribute work	full context + request	order-B, order-C	—
B	change code	order-B + target code	code	—
C	write tests → run → verdict	order-C	tests, result	—
D	analyze → surface limitations, update docs	order + result + context	full context	✓
E	refactor + update architecture doc	code + existing tests + devlog	code, 03-architecture	✓

Only A reads the full context. It's the most expensive and most important step, so tokens are concentrated there while B · C are kept narrow.

Permission boundaries

- B cannot modify test code. Otherwise it bends the code to fit the tests.
- Only C may modify tests. C also gives the verdict.
- E doesn't touch 01-features · 02-design.
- Only D, E and F may commit, and only when commit is on (see Commit mode).

Main loop



C only needs A's acceptance criteria to write tests, so it doesn't wait for B. Tests standing first is what later lets E refactor safely.

A starts the loop only when it's confident. If the intent of a request isn't clear right away, it doesn't write orders and asks the user first. This isn't a design approval gate — it doesn't seek approval of a design, it confirms what is wanted, and once confirmed the loop runs to the end without human intervention.

There are three signals of ambiguity: the request branches into several designs, it conflicts with the existing spec, or its scope can't be written in one line.

Retry constraints

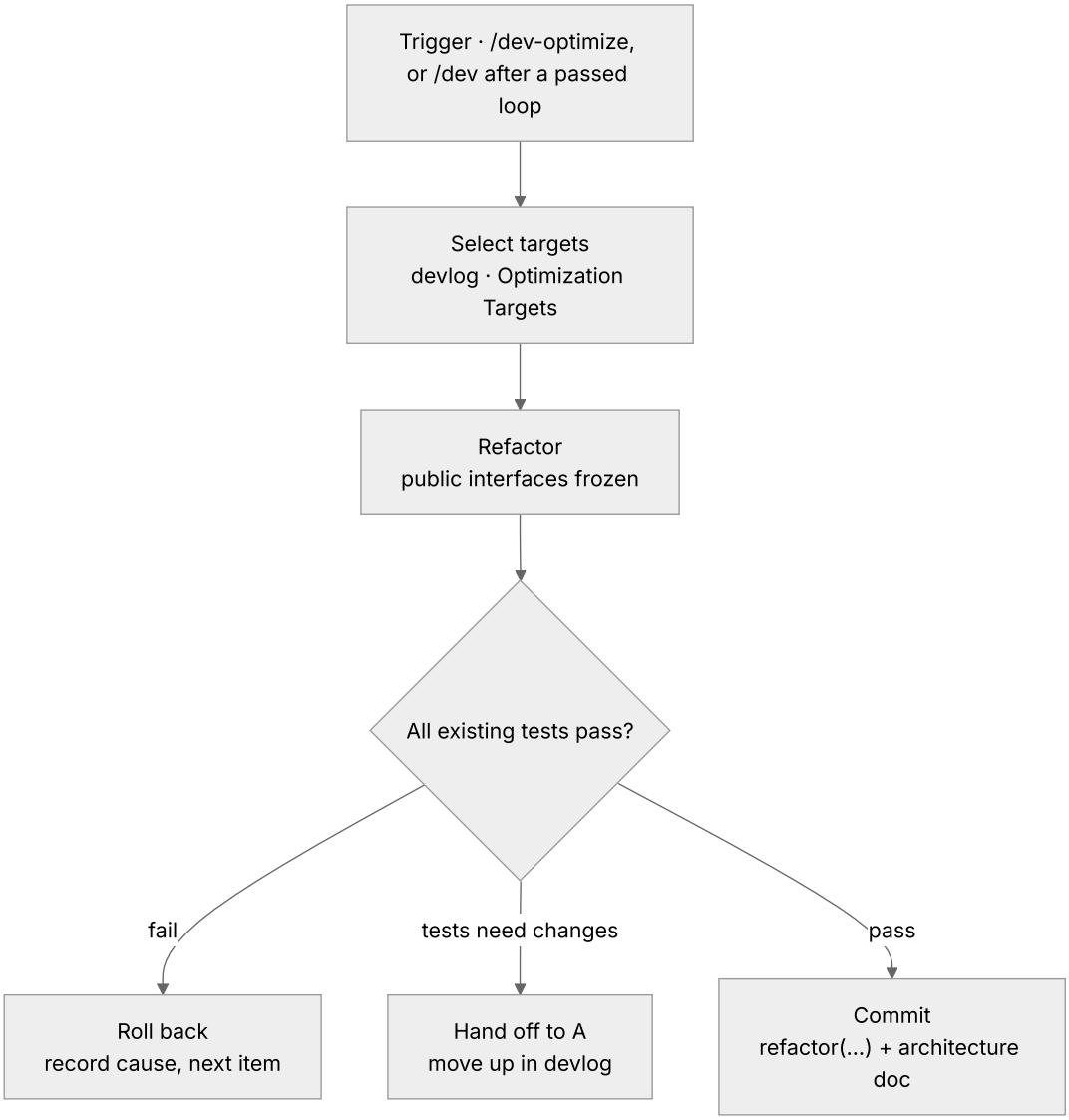
Six limits keep the B↔C loop from spinning forever or burning tokens.

1. **Max 3 tries.** Stop when exhausted.

2. **Failures are passed as summaries only.** Failing test name + assertion diff + file:line. Never pass full logs.
3. **Same failure twice in a row → stop at once.** Circling the same spot isn't solved by retrying.
4. **B can't touch test code.** Only C may modify tests.
5. **C gives the verdict.** Tests come from A's acceptance criteria, so by default the tests are right.
6. **No commit when exhausted.** D records the cause of failure in 04-devlog, reports, and ends. The failure becomes context too, and A redesigns in the next loop.

E · Optimization loop

E runs after a loop has passed and been recorded — never concurrently with one (they touch the same code, so they're serialized).



One rule: nothing observable from the outside changes.

E does	E doesn't
remove duplication	implement missing features
break up long functions · classes	add missing error handling
remove dead code	fix bugs
make naming consistent	change public API signatures
clean up unneeded dependencies	optimizations that change behavior, like adding a cache
remove redundant computation · N+1	preemptive abstraction for extensibility

There is one test. **If existing tests don't change by a single line, it's E; if they have to, it's not E.**

- No limit on the number of targets. Several optimization targets recorded in the devlog can be handled in one run. Unrecorded wholesale refactoring is still forbidden.
- No retries. Roll back immediately on failure.
- If it seems 01-features · 02-design need changes, it's out of scope.
- Triggered by `/dev-optimize`, or **automatically** by `/dev` after a passed loop when `devtool.py auto` reports 3+ targets or one marked `[high]` (complexity that already gets in the way: a function over ~150 lines, the same logic in 3+ places, a module with 4+ responsibilities). D only records targets; the orchestrator decides from the count, so the trigger doesn't depend on judgment.

F · Memory compaction

A and D read all of `.dev/context/` on every loop, so the docs' size is paid on every request, and it grows with the project. F rewrites them shorter **without changing their meaning**: prose becomes terse bullets, repeated explanations merge, resolved text that is already superseded goes, and Loop Log rows beyond the latest 20 move verbatim to `.dev/archive/loop-log.md` (not read by the loop). It never drops or renumbers a spec ID, never merges two open devlog items, and keeps every heading.

`devtool.py memory-check` enforces this against HEAD — every spec ID (in context or archive), every heading and the open-item counts must survive, or F restores them or rolls back. The size after compaction is recorded in `.dev/memory.json`, and `/dev` runs F again only when the docs are over budget (~6000 tokens) **and** 25% larger than after the last compaction (or the Loop Log passes 20 rows) — so it doesn't churn on every loop. `/dev-optimize-memory` runs it on demand.

Commit convention

One loop = one commit. Code + tests + docs are one unit. E makes one commit per item it handles, separated with the `refactor(...)` prefix.

```
feat(auth): add social login

Intent: reduce drop-off at sign-up (request: "add social login")
Design: abstracted behind an AuthProvider interface, extensible to OAuth providers
Changes: F-031 new / D-12 modified / U-08 new
Tests: 7 added, all passing
Limits: token refresh not implemented → devlog#14

loop: 2026-10-21-001
```

Pinning A's design intent in the message is the key. With the approval gate removed, this is the only place to trace afterwards "what A intended".

Safeguards

There is no design approval gate. Instead, these safeguards fill its place.

Safeguard	What it does
Ask when intent is unclear	If A isn't confident, it talks with the user first instead of writing orders
Stop on destructive changes	Schema migrations, mass deletions and major dependency upgrades stop and report
Loop serialization	With commit on, loops start only on a clean git working tree. Keeps E and the main loop from editing the same files at once
Match check	At the end of every loop, D leaves one line on "does the implementation match the request"
Design intent record	A's intent and the loop ID go into the commit message so the point of divergence can be traced

When a mismatch is detected, A reads it first on the next request. The line of defense moves from before to after, but stays inside the loop.

Commit mode

Commits are optional. `.dev/config.json` holds `"commit": "auto" | "on" | "off"`; `auto` (the default) commits **only in a git repo that has a remote** — a project not connected to GitHub (no git, or a local repo without a remote) runs with commit off.

Git did three jobs in the loop, so commit off replaces each:

Job	Commit on	Commit off
History	one commit per loop / per E item	the devlog's Loop Log only
Serialization	clean working tree before a loop	not enforced — don't run two loops at once
Rollback (E, F) and F's no-loss check	`git checkout` / compare with `HEAD`	`devtool.py snapshot` before, `restore` on failure; `memory-check` compares with the snapshot

Rollback never uses git when commit is off, because the tree may hold uncommitted work that `git checkout` would destroy.

Automation mapping

Mechanism	What it's given
CLAUDE.md	only what must hold in every session: when the loop runs, the language rule, a pointer to the skill
Skill	loop rules (doc boundaries, permissions, retries, stop conditions, commit format, housekeeping), spec templates, doc update checklist
Subagent	each of A–F — the implementation of context independence
Hook	enforce tests before commit, load 04-devlog at session start
Script	`devtool.py` — preflight and commit mode, snapshot/restore, devlog sections, loop pruning, running tests with condensed output, and the numbers behind automatic E / F (`auto` , `memory` , `memory-check`)
Command	`/dev` , `/dev-optimize` , `/dev-optimize-memory` , `/dev-limitations`

Principle: always-needed rules in CLAUDE.md (kept short — it is read in every session and by every subagent), loop rules and procedures in skills, enforcement in hooks, isolation in subagents.

Two layers: the global level lays down only the loop structure, and each project's CLAUDE.md fills in the specifics (test command, directory locations, language · framework). If a project has no CLAUDE.md, A scans the repo on the first run and generates one automatically.

Caveats when adopting

The first 3–5 loops are not hands-off. If `01-03` are empty, A has no context to read and rescans all the code every time. For an existing project, it's better to spend the first loop on "documenting the current state in reverse".

A being quietly wrong is the only real risk. Mistakes by B · C · D show up in tests or diffs, but if A misreads the intent you get "a perfectly implemented wrong feature", and the tests pass too.

Settled decisions

Item	Decision
`loops/` retention	last 10, the rest deleted automatically
Project CLAUDE.md	if missing, A scans the repo and generates it on the first run
Loop serialization	enforced by a clean git working tree check when commit is on (no separate lock file)
Commits	`.dev/config.json` `commit`: `auto` (default — only with a git remote), `on`, `off`
E trigger	`/dev-optimize`, or automatic after a passed loop at 3+ targets or one `[high]`
F trigger	`/dev-optimize-memory`, or automatic after a passed loop when the context is over budget and has grown 25% since the last compaction